QUANT RESEARCH / STATISTICS

ALPHA VALIDITY

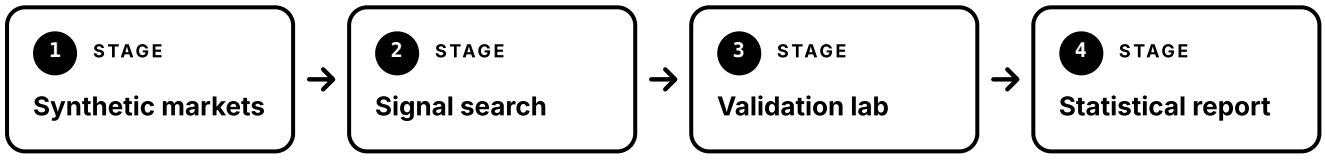
PROJECT 07

Scientific scepticism for quantitative strategy research.

A scientific investigation into false discoveries in strategy research, measuring how selection bias, overfitting and costs distort apparent alpha.

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM / RESEARCH ARCHITECTURE



THE PROBLEM TO SOLVE

How well do common validation methods distinguish a real predictive signal from a chance pattern?

TECHNICAL DISTINCTION

1. The point is methodological honesty, not presenting a flashy equity curve.
2. Strong negative results can be as impressive as a profitable toy backtest if the methodology is rigorous.

WHAT THE EVIDENCE MUST SHOW

- Empirical false discovery rate and statistical power
- Sensitivity to regime shift and data snooping
- Synthetic market conclusions versus licensed historical data